MORPHISMS OF SIMPLICIAL SITES LIFT TO MORPHISMS OF RINGED TOTAL TOPOI WITH COMPONENTWISE PUSHFORWARD COMPATIBILITY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: companion of card L46154 (morphism-simplicial-sites, sheaf version) and card L46168 (restriction-module-to-components-site); the 'small point' of the author (source lines 1637-1643) is the definition of h_n^sharp, with the required identifications g_n^{-1}h_{total}^{-1}O' = h_n^{-1}(g'_n)^{-1}O' = h_n^{-1}O'_n spelled out; no verification is omitted.

Editorial note. The label lemma-morphism-simplicial-sites-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation and ringed morphisms of simplicial sites. Same chapter, Situation \label{situation-simplicial-site} and Remark \label{remark-morphism-simplicial-sites-modules}. Statement lines 1604-1612: O, O' sheaves of rings on the total sites; (h, h^sharp) a morphism between simplicial sites as in the remark.
- The sheaf-level morphism of total topoi. Same chapter, \label{lemma-morphism-simplicial-sites} (card L46154). Proof lines 1635-1637: the morphism of topoi h_total and the commutative diagrams at the sheaf level, from which the ringed version follows.
- The ringed component restrictions g_n. Same chapter, \label{lemma-restriction-module-to-components-site} (card L46168). Statement lines 1629-1631 and proof lines 1637-1644: g_n and g'_n as morphisms of ringed topoi; the definition of h_n^sharp uses their ring identifications.

Source reference. Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 1637; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 1629; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DGZ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1588–1588. Prerequisite pointer invoked at proof line(s) 1611; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1607; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}_n, f_\varphi, u_\varphi$ and $\mathcal{C}'_n, f'_\varphi, u'_\varphi$ be as in Situation 09WE. Let $\mathcal{O}$ and $\mathcal{O}'$ be a sheaf of rings on $\mathcal{C}_{total}$ and $\mathcal{C}'_{total}$. Let $(h, h^\sharp)$ be a morphism between simplicial sites as in Remark 0DGZ. Then we obtain a morphism of ringed topoi*

$$h_{total} : (Sh(\mathcal{C}_{total}), \mathcal{O}) \to (Sh(\mathcal{C}'_{total}), \mathcal{O}')$$

and commutative diagrams

$$\begin{array}{ccc} (Sh(\mathcal{C}_n), \mathcal{O}_n) & \xrightarrow{h_n} & (Sh(\mathcal{C}'_n), \mathcal{O}'_n) \\ \downarrow{\scriptstyle g_n} & & \downarrow{\scriptstyle g'_n} \\ (Sh(\mathcal{C}_{total}), \mathcal{O}) & \xrightarrow{h_{total}} & (Sh(\mathcal{C}'_{total}), \mathcal{O}') \end{array}$$

of ringed topoi where g_n and g'_n are as in Lemma 0D72. Moreover, we have $(g'_n)^* \circ h_{total,*} = h_{n,*} \circ g_n^*$ as functor $Mod(\mathcal{O}) \to Mod(\mathcal{O}'_n)$.

Proof. Follows from Lemma 0D96 and 0D72 by keeping track of the sheaves of rings. A small point is that in order to define h_n as a morphism of ringed topoi we set $h_n^\sharp = g_n^{-1} h^\sharp : g_n^{-1} h_{total}^{-1} \mathcal{O}' \to g_n^{-1}\mathcal{O}$ which makes sense because $g_n^{-1} h_{total}^{-1} \mathcal{O}' = h_n^{-1}(g'_n)^{-1}\mathcal{O}' = h_n^{-1}\mathcal{O}'_n$ and $g_n^{-1}\mathcal{O} = \mathcal{O}_n$. Note that $g_n^*\mathcal{F} = g_n^{-1}\mathcal{F}$ for a sheaf of $\mathcal{O}$-modules $\mathcal{F}$ and similarly for g'_n and this helps explain why $(g'_n)^* \circ h_{total,*} = h_{n,*} \circ g_n^*$ follows from the corresponding statement of Lemma 0D96. $\square$